Lecture 5: Linear Models for Classification

Tao LIN

SoE, Westlake University

March 18, 2025

1. Review of Last Week
   - Data Model and Learning Algorithm
   - Generalization Gap and Model Selection
   - Bias-Variance Decomposition

2. Logistic Regression
   - Classification
   - Logistic Regression

Reading materials & Reference

Reading materials:

- Chapter 3, Stanford CS 229 Lecture Notes, `https://cs229.stanford.edu/notes2022fall/main_notes.pdf`
- Lecture 4, Stanford CS 231n, `http://cs231n.stanford.edu/schedule.html`

Reference:

- EPFL, CS-433 Machine Learning, `https://github.com/epfml/ML_course`

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How do we trade-off the under-fitting and over-fitting?

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Data model:

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- $\mathcal{A}$ denotes the learning algorithm.
- S denotes the input (dataset).

True Error, Empirical Error, and Training Error

- **True Error** (the **expected error** over all samples chosen according to $\mathcal{D}$):

$$\mathcal{L}_{\mathcal{D}}(f) = \mathbb{E}_{\mathcal{D}}\left[\ell(y, f(\mathbf{x}))\right] . \tag{3}$$

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The reason that $L_S(f_S)$ might not be close $L_{\mathcal{D}}(f_S)$ is of course over-fitting.

Problem: validating the model on the same data subset we trained it on!

Splitting the data and Test Error

- **Fix:** Split the data into a *training* set S_{train} and a *test* set S_{test} (a.k.a. *validation* set):

$$S = S_{\text{train}} \bigcup S_{\text{test}} \tag{6}$$

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- Issues: we have fewer data both for the learning and validation tasks (trade-off)

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The variation of the $\left|L_{\mathcal{D}}(f) - L_{S_{\text{test}}}(f)\right|$ matters.

Theorem 1

Given a model f and a test set $S_{test} \sim \mathcal{D}$ i.i.d. (not used to learn f) and a loss $\ell(\cdot,\cdot) \in [a,b]$:

$$\Pr\left[\left|L_{\mathcal{D}}(f) - L_{S_{test}}(f)\right| \geq \sqrt{\frac{(b-a)^2 \ln(2/\delta)}{2\left|S_{test}\right|}}\right] \leq \delta \tag{12}$$

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Given a predictor f and a dataset S, we can control the expected risk:

$$\Pr\left[\underbrace{L_{\mathcal{D}}(f)}_{\text{not computable}} \geq \underbrace{L_{S_{\text{test}}}(f)}_{\text{computable}} + \underbrace{\sqrt{\frac{(b-a)^2 \ln(2/\delta)}{2\,|S_{\text{test}}|}}}_{\text{deviation}}\right] \leq \delta\,. \tag{13}$$

How far is each of the K test errors $L_{S_{\text{test}}}(f_k)$ from the true $L_{\mathcal{D}}(f_k)$?

Theorem 2

We can bound the maximum deviation for all K candidates, by

$$\Pr\left[\max_k |L_{\mathcal{D}}(f_k) - L_{S_{test}}(f_k)| \geq \sqrt{\frac{(b-a)^2 \ln(2K/\delta)}{2\,|S_{test}|}}\right] \leq \delta \tag{14}$$

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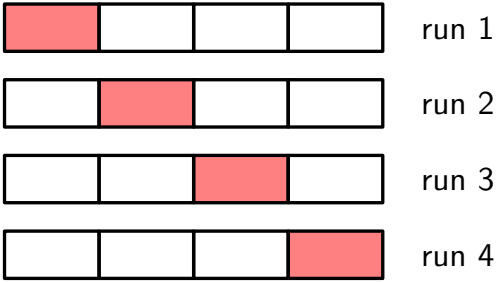
$\Rightarrow$ So we can test many different models without incurring a large penalty.

Issues: Splitting the data once into two parts (one for training and one for testing) is not the most efficient way to use the data!

K-fold cross-validation:

1. Randomly partition the data into K groups
2. Train K times. Each time leave out exactly one of the K groups for testing and use the remaining $K-1$ groups for training.
3. Average the K results



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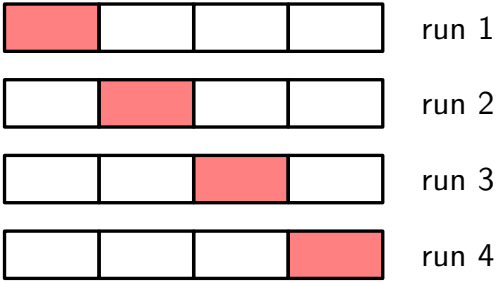

Benefits:

- We have used all data for training, and all data for testing, and used each data point the same number of times.

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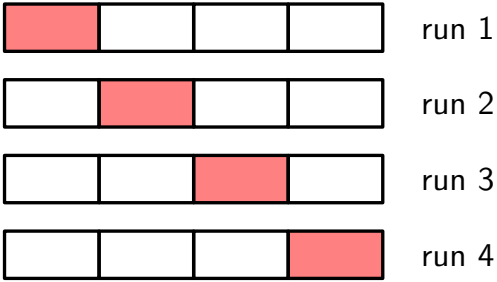

Benefits:

- We have used all data for training, and all data for testing, and used each data point the same number of times.
- Cross-validation returns an unbiased estimate of the generalization error and its variance.

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Bias-Variance Decomposition

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&+ \left(f(\mathbf{x}_0) - \mathbb{E}_{S_{\text{train}'}}\left[f_{S_{\text{train}'}}(\mathbf{x}_0)\right]\right)^2 && \text{(Bias)} \\
&+ \mathbb{E}_{S_{\text{train}} \sim \mathcal{D}}\left[\left(\mathbb{E}_{S_{\text{train}'}}\left[f_{S_{\text{train}'}}(\mathbf{x}_0)\right] - f_{S_{\text{train}}}(\mathbf{x}_0)\right)^2\right], && \text{(Variance)}
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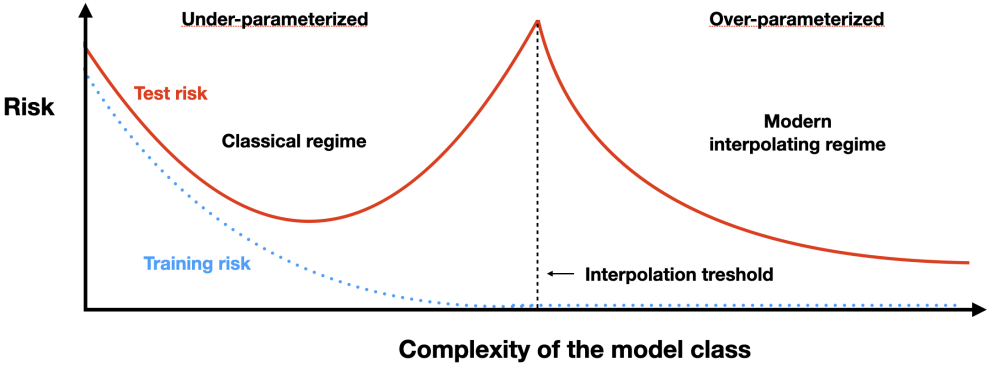
$\Rightarrow$ To minimize the true error, we need to select a method that **simultaneously achieves low bias and low variance**.

Double descent curve in Deep Learning



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This lecture:

- Logistic Regression
- Exponential Families and Generalized Linear Models

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<u>How:</u> relates input to a categorical variable

Classifier

A classifier $f : \mathcal{X} \to \mathcal{Y}$

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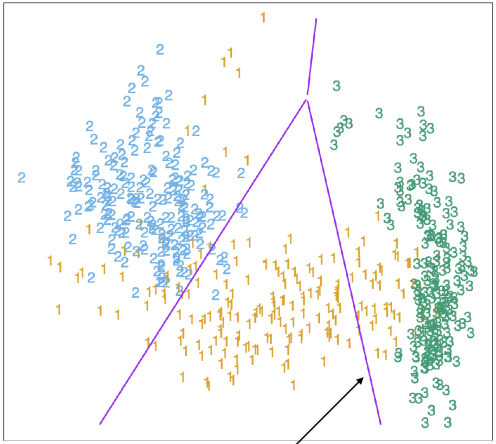

Linear Decision boundary

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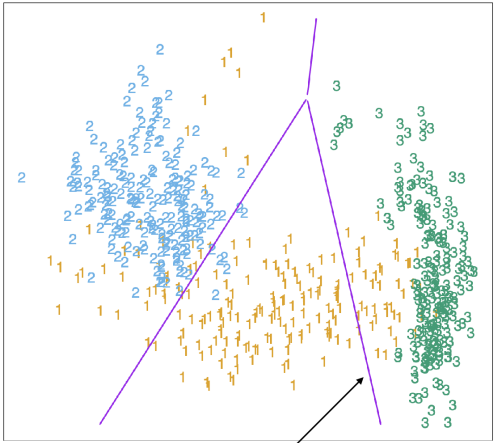

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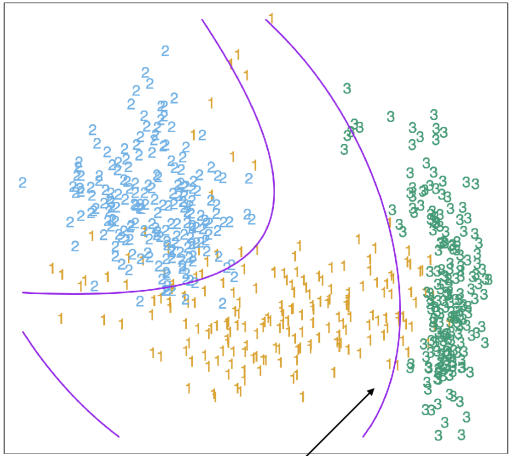

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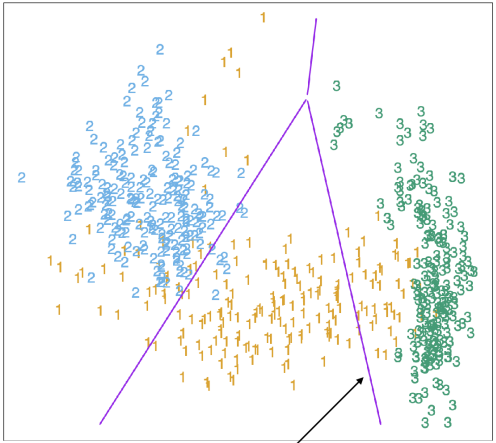

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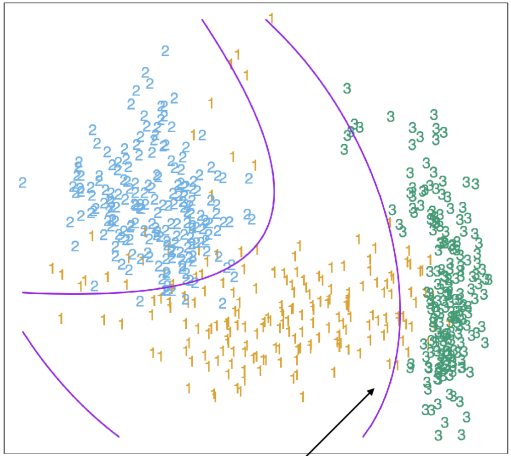

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Classification: a special case of regression?

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$$(\mathbf{x}, y) \in \mathcal{X} \times \{0, 1\} \subset \mathcal{X} \times \mathbb{R} \tag{15}$$

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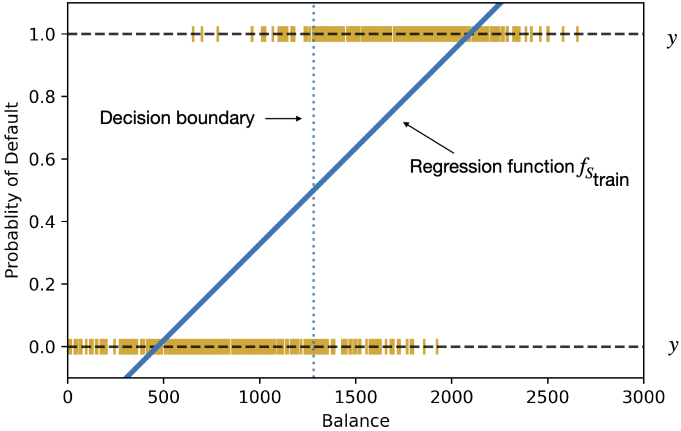
Could we use previously seen regression methods to solve it?

Is it a good idea to use some regression methods?



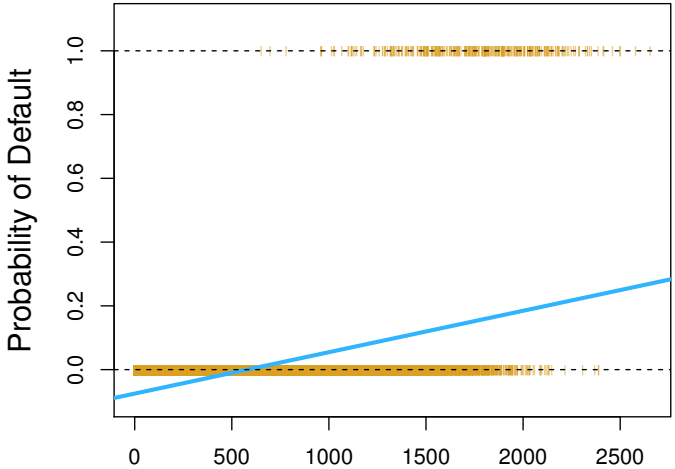
$y = 1$ **if "default"**

$y = 0$ **if "no default"**

We label the output as probability for sake of interpretation

Classification is not just a special form of regression

- The predicted values are not probabilities (not in $[0, 1]$)



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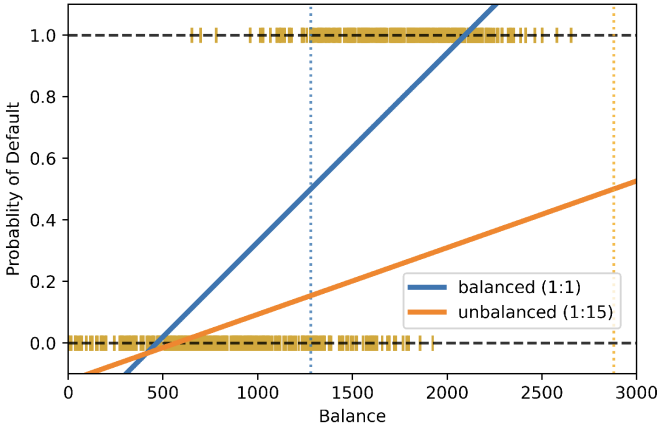
- Sensitivity to unbalanced data



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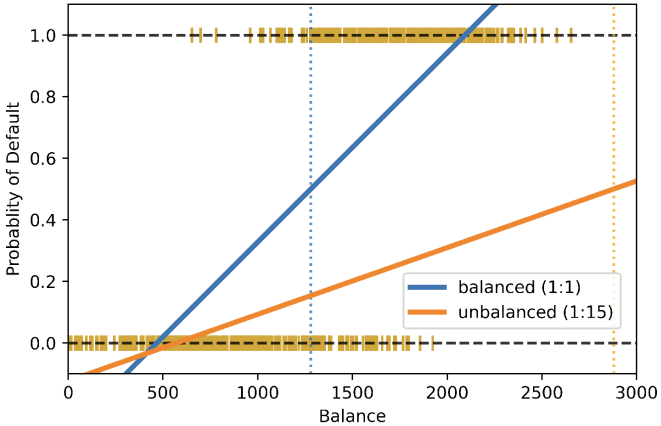

The position of the line depends crucially on how many points are in each class.

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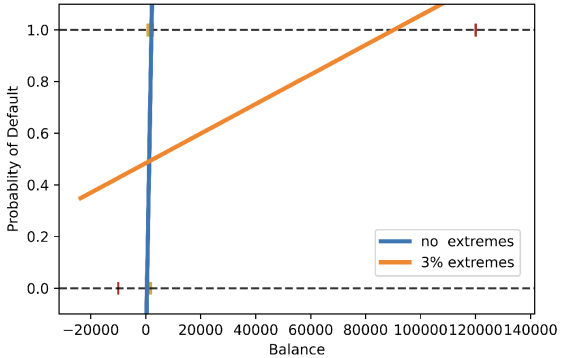

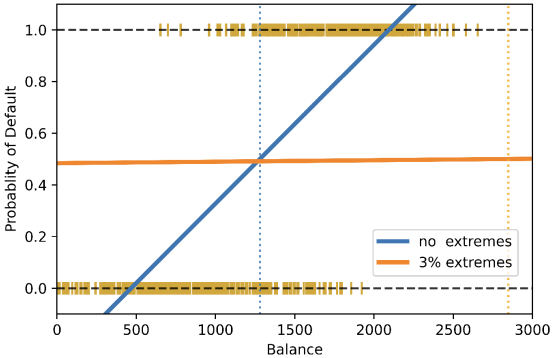

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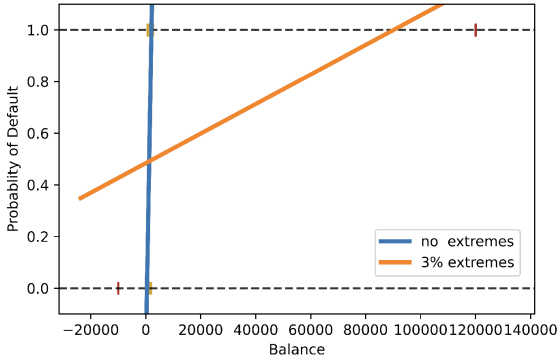

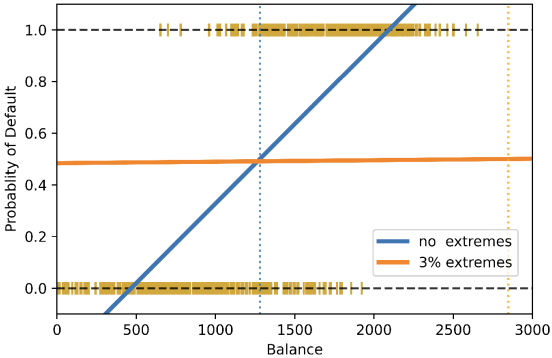

The position of the line depends crucially on where the points lie.

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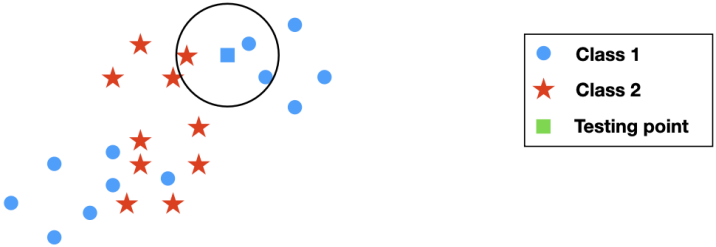
- Fundamental task of classification:

Divide the space into distinct decision regions

k-Nearest Neighbor

Assume that nearby points are likely to have similar labels



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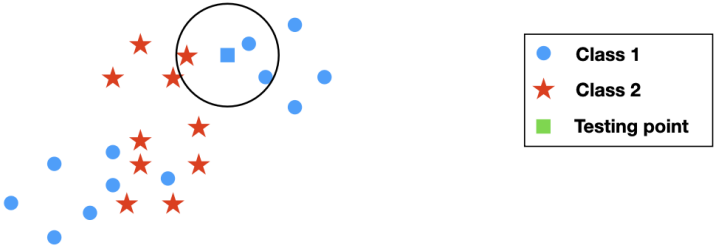

A new point $\mathbf{x}$ is classified based on the **majority vote of its k-nearest neighbors**.

k-Nearest Neighbor

Pros:

Cons:

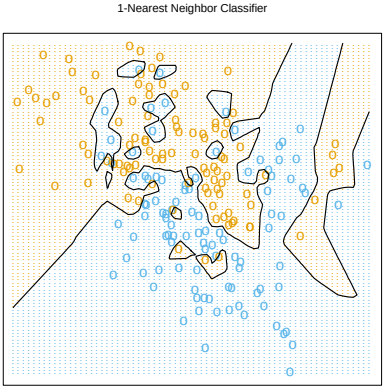

FIGURE 2.3. *The same classification example in two dimensions as in Figure 2.1. The classes are coded as a binary variable* (BLUE $= 0$, ORANGE $= 1$), *and then predicted by* 1*-nearest-neighbor classification.*

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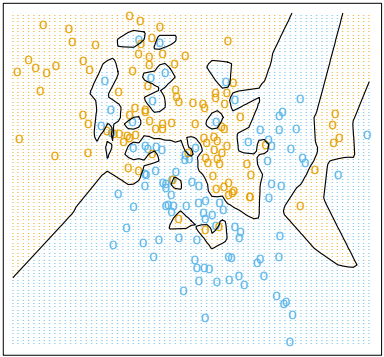

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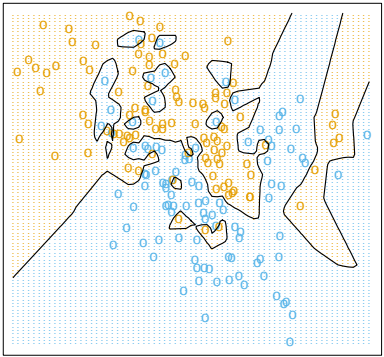

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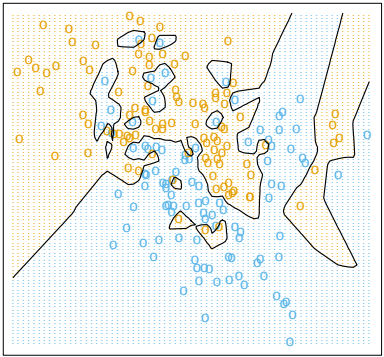

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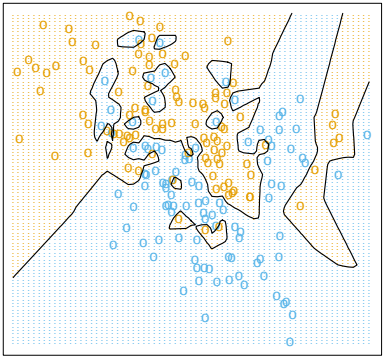

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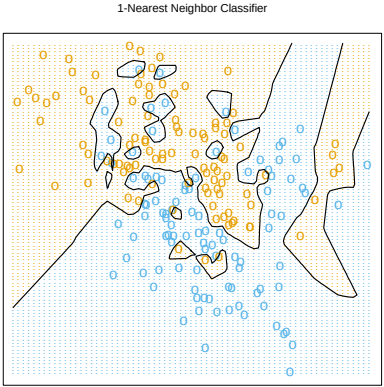

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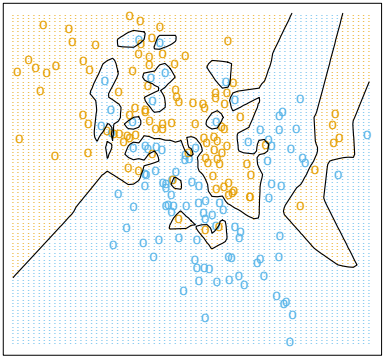

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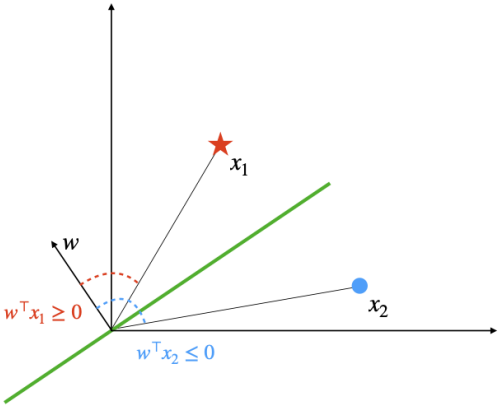

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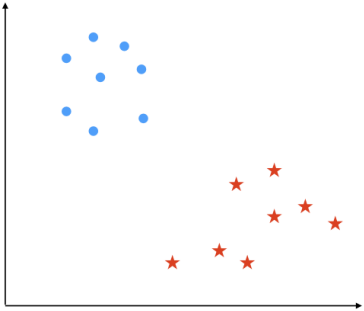

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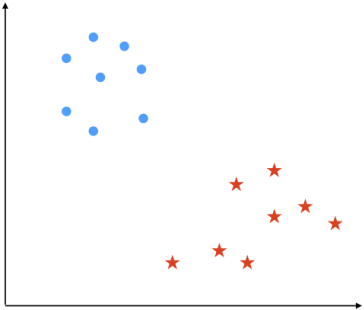

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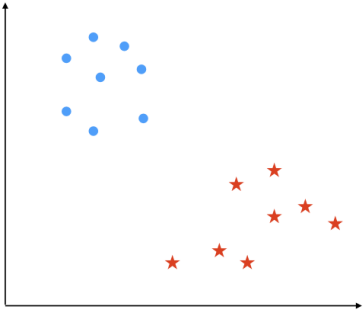
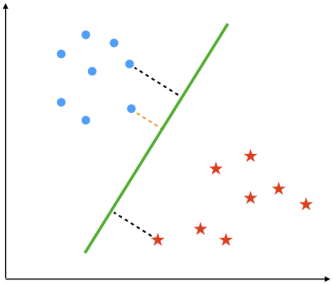


Assume the data are linearly separable, i.e., a separating hyperplane exists

Which separating hyperplane would you pick?

Margin

Key concept: The margin is the distance from the hyperplane to the closest point.



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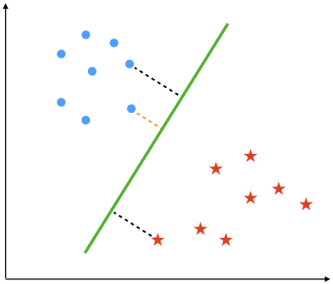
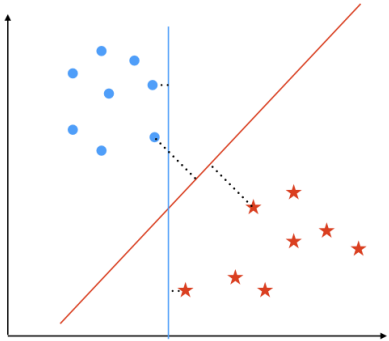


Take the one with the largest margin!

Max-margin separating hyperplane

Choose the hyperplane which maximizes the margin



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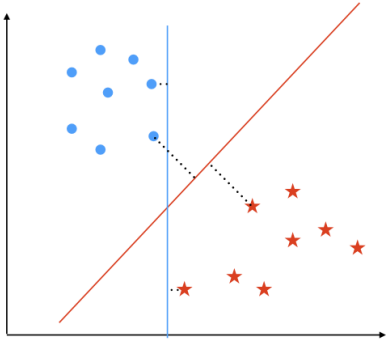

WHY?

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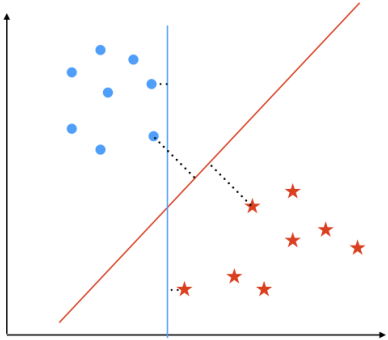

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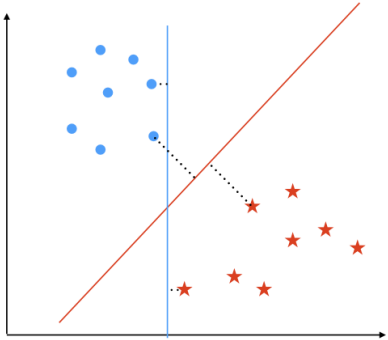
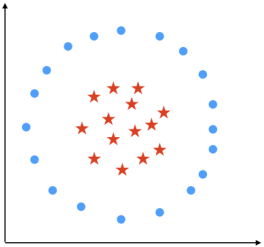


WHY? If we slightly change the training set, the number of misclassifications will stay low

$\Rightarrow$ It will lead us to support vector machine (SVM) and logistic regression!

Nonlinear classifier



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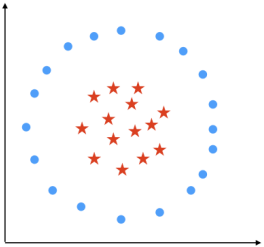

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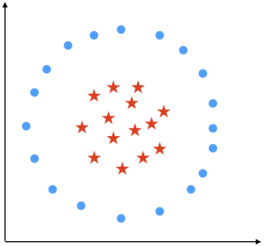

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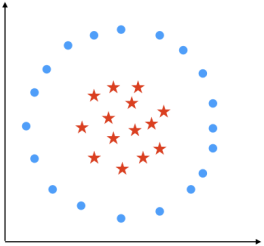

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- Goal: minimize $L_{\mathcal{D}}(f)$

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$$f^\star(\mathbf{x}) \in \arg\max_{\{-1,1\}} \Pr(Y = y|X = \mathbf{x}) \tag{17}$$

- In practice, we do not know the joint distribution $p(\mathbf{x}, y)$
  $\Rightarrow$ Bayes classifier is an unattainable gold standard.
  But we can use the data to learn the distribution (by assuming the data distribution).

Proof of the Bayes classifier

Claim 1: $\forall \mathbf{x} \in \mathcal{X}, f(\mathbf{x}^\star) \in \arg\min_{y \in \mathcal{Y}} \Pr(Y \neq y | X = \mathbf{x}) \quad \Longrightarrow \quad f^\star \in \arg\min_{f:\mathcal{X}\to\mathcal{Y}} L_{\mathcal{D}}(f)$

$$L_{\mathcal{D}}(f) = \mathbb{E}_{X,Y}\left[1_{Y \neq f(X)}\right] = \mathbb{E}_X\left[\mathbb{E}_{Y|X}\left[1_{Y \neq X}|X\right]\right] \tag{18}$$

$$= \mathbb{E}_X\left[\Pr(Y \neq f(X)|X)\right] \tag{19}$$

$$= \mathbb{E}_X\left[\min_{y \in \mathcal{Y}} \Pr(Y \neq y|X)\right] \tag{20}$$

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Claim 2: $f^\star(\mathbf{x}) \in \arg\min_{y \in \mathcal{Y}} \Pr(Y \neq y | X = \mathbf{x})$

$$f^\star(\mathbf{x}) \in \arg\max_{y \in \mathcal{Y}} \Pr(Y = y | X = \mathbf{x}) = \arg\min_{y \in \mathcal{Y}} \Pr(Y \neq y | X = \mathbf{x}) \tag{22}$$

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- **Parametric:** approximate true distribution $\mathcal{D}$ via training data
  $\Rightarrow$ Minimize the empirical risk on training data (ERM)

Recap

- Classification:
  - Mapping inputs to discrete outputs (categorical)
  - Not a special form of regression!
- Ways to perform classification:
  - **Non-parametric**: K-Nearest-Neighbors
  - **Parametric**: learning X-to-Y mapping via ERM
  - More complex decision boundaries? Non-linear classifiers

Table of Contents

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- Data Model and Learning Algorithm
- Generalization Gap and Model Selection
- Bias-Variance Decomposition

2 Logistic Regression
- Classification
- Logistic Regression

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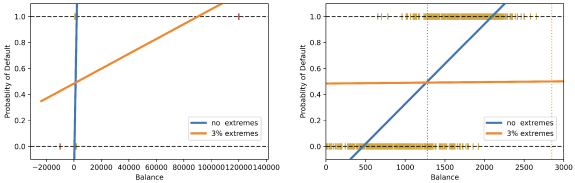

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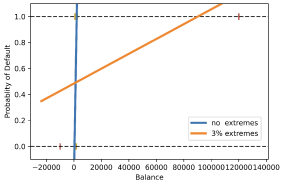
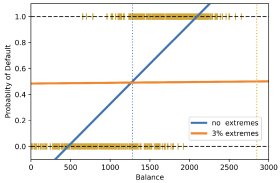
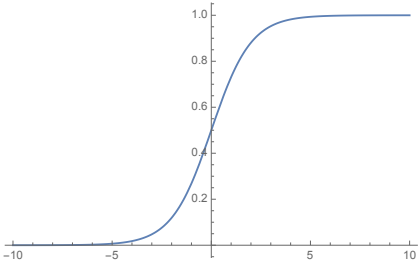




Solution: Transforming the predictions that take values in $(-\infty, \infty)$ into $[0, 1]$.

The logistic function

Consider first of all the case of two classes.



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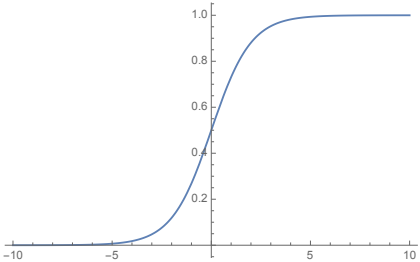

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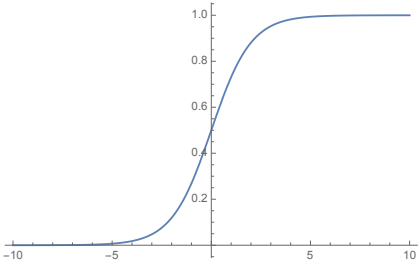

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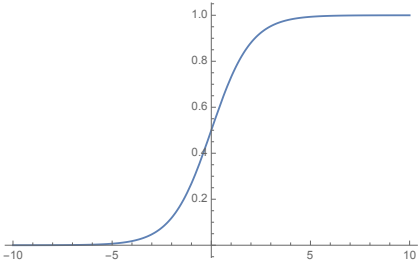

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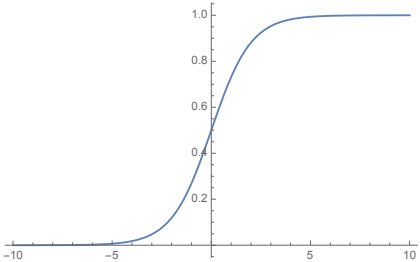

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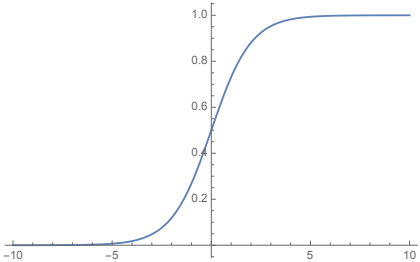

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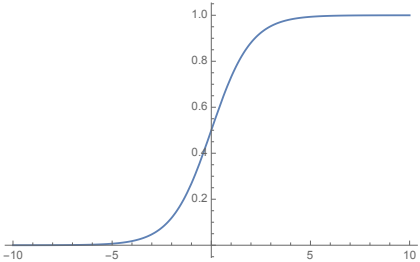

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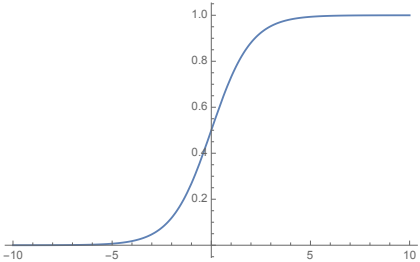

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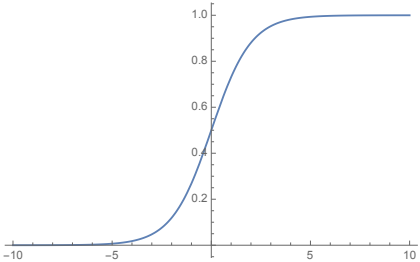

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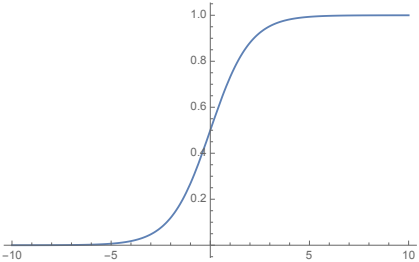

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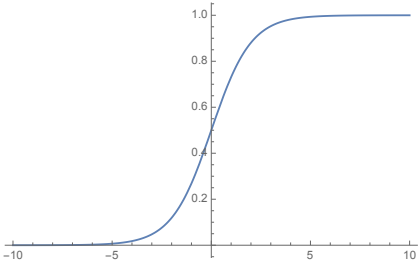

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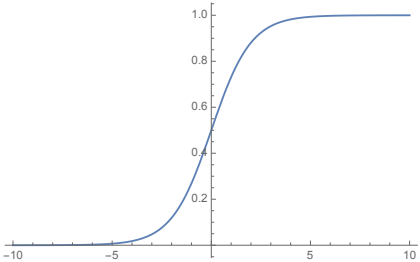

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Interpretation:

- Very large $\mathbf{x}^\top \mathbf{w} + w_0$ corresponds to $p(1|\mathbf{x})$ very close to 0 or 1 (high confidence).

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- Very large $\mathbf{x}^\top \mathbf{w} + w_0$ corresponds to $p(1|\mathbf{x})$ very close to 0 or 1 (high confidence).
- Small $\left|\mathbf{x}^\top \mathbf{w} + w_0\right|$ corresponds to $p(1|\mathbf{x})$ very close to 0.5 (low confidence).

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the MLE converges to the true parameter when $N \to \infty$.

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As a result,

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Gradient of the negative log likelihood

Recall that

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$\Rightarrow$ Good news: the cost function L is convex.

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Second proof: Hessian of $\mathcal{L}$ is psd

The Hessian $\nabla^2\mathcal{L}$ is the **matrix** whose entries are the **second derivatives** $\frac{\partial^2}{\partial w_i \partial w_j}\mathcal{L}(\mathbf{w})$.

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\end{aligned}$$

It can be written under the matrix form:

$$\nabla^2\mathcal{L}(\mathbf{w}) = \frac{1}{N}\mathbf{X}^\top\mathbf{S}\mathbf{X}\,, \qquad \text{where } \mathbf{S} = \operatorname{diag}\left[\sigma(\mathbf{x}_n^\top\mathbf{w})\left(1-\sigma(\mathbf{x}_n^\top\mathbf{w})\right)\right] \succeq 0 \tag{44}$$

$\Longrightarrow$ $\mathcal{L}$ is convex since $\nabla^2\mathcal{L}(\mathbf{w}) \succeq 0$.